
PROJECT EXPO

TEAM MEMBERS:

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SMART IoT-BASED SHOPPING TROLLEY USING RFID

BRAINSTORMING REPORT

1. Problem Areas to Address

- Long queues at billing counters.
- Lack of real-time bill visibility for customers.
- Difficulty in budgeting during shopping.
- No in-store navigation assistance.
- Manual inventory tracking errors.
- Accessibility challenges for elderly/differently-abled customers.
- High staffing costs during peak hours.

2. Idea Generation (Possible Solutions)

- **RFID-Based Billing:** Automatic detection of items placed in trolley.
- **Real-Time Bill Display:** LCD/LED screen showing updated bill continuously.
- **Cloud Integration:** Sync trolley data with store inventory for real-time stock updates.
- **Digital Payment Integration:** Enable contactless payments directly from trolley.
- **Exit Verification System:** Cross-check trolley items with billed list at store exit.

- **In-Store Navigation:** BLE/RFID zone mapping to guide customers to product sections.
- **Accessibility Features:** Lightweight trolley design, voice assistance, easy interface.
- **Data Analytics:** Collect shopping patterns for personalized recommendations.
- **Energy Efficiency:** Low-power microcontroller, rechargeable battery system.
- **Scalability:** Modular design adaptable for small and large stores.

3. Grouping of Ideas

- **Customer Experience**
 - Real-time bill display
 - Navigation assistance
 - Accessibility features
- **Store Management**
 - Inventory sync
 - Data analytics
- **Technology & Infrastructure**
 - RFID integration
 - Cloud database
 - Energy-efficient hardware
- **Operational Efficiency**
 - Digital payment
 - Reduced staffing needs
 - Faster checkout

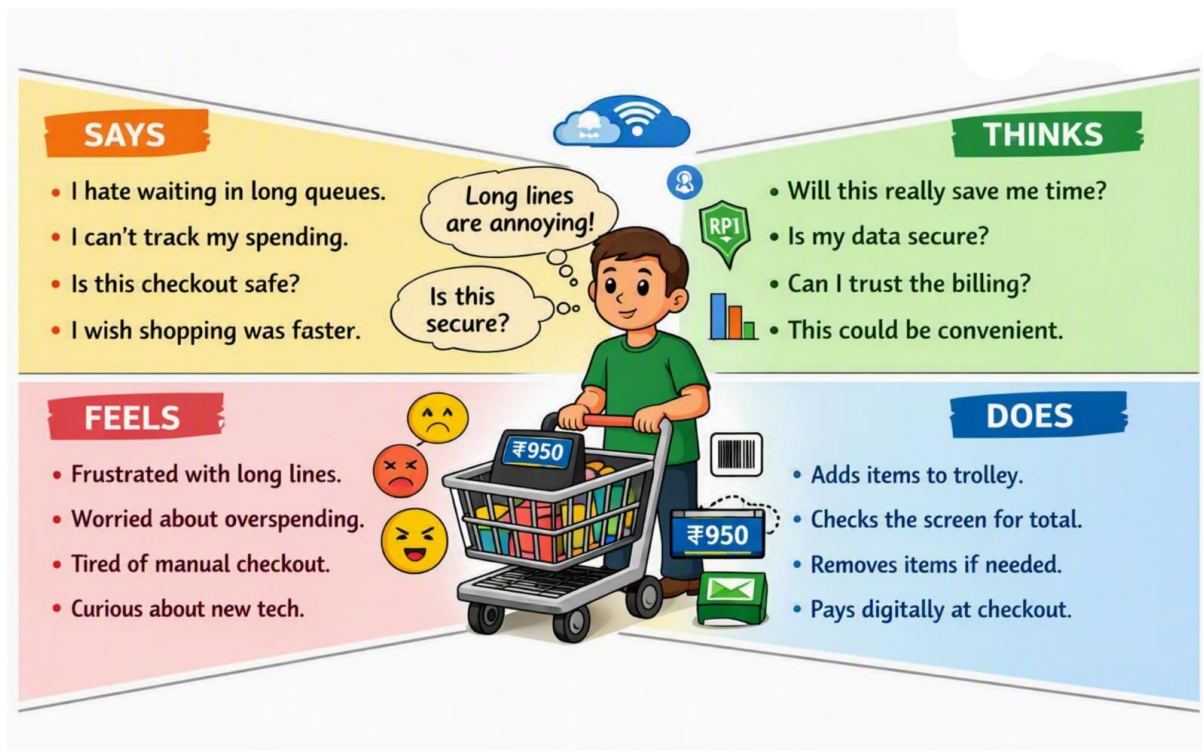
4. Early Insights

- RFID-based billing is the **core enabler** of the system.
- Navigation and accessibility features add **extra value** but may be phased in later.
- Cost and energy efficiency are **key for scalability** in small/medium retailers.

IDEA PRIORITIZATION MATRIX

Idea / Feature	Impact (High/Med/Low)	Effort (High/Med/Low)	Priority (Result)
RFID-Based Automatic Billing	High	Medium	★ Top Priority
Real-Time Bill Display	High	Low	★ Top Priority
Cloud Inventory Sync	High	Medium	★ Top Priority
Digital Payment Integration	High	High	Medium Priority
Exit Theft Verification System	High	High	Medium Priority
In-Store Navigation Assistance	Medium	High	Low Priority (Future Scope)
Accessibility Features (voice, lightweight trolley)	Medium	Medium	Medium Priority
Data Analytics (shopping patterns)	Medium	Medium	Medium Priority
Energy-Efficient Hardware	Medium	Low	★ Quick Win
Scalability for Small Stores	High	Medium	★ Top Priority

EMPATHY MAP



USER PERSONA

Persona 1: Lakshmi – Busy Homemaker

- **Age:** 38
- **Occupation:** Teacher
- **Shopping Behaviour:** Weekly grocery shopping for family of 4.
- **Goals:**

- Save time at checkout.
- Track expenses while shopping.
- **Pain Points:**
 - Long queues at billing counters.
 - Difficulty managing budget without real-time bill visibility.
- **Needs:**
 - Easy-to-use trolley interface.
 - Real-time bill display.
 - Digital payment option.

Persona 2: Ramesh – Elderly Shopper

- **Age:** 65
- **Occupation:** Retired bank employee
- **Shopping Behaviour:** Visits supermarket twice a week for essentials.
- **Goals:**
 - Simple, stress-free shopping experience.
 - Avoid heavy lifting and complex technology.
- **Pain Points:**
 - Difficulty handling manual scanning.
 - Struggles with heavy trolleys.
- **Needs:**
 - Lightweight trolley design.
 - Voice-assisted interface.
 - Automatic billing without manual scanning.

Persona 3: Priya – Tech-Savvy Millennial

- **Age:** 27
- **Occupation:** Software Engineer
- **Shopping Behaviour:** Prefers quick, efficient shopping; uses digital wallets.
- **Goals:**

- Seamless checkout with contactless payment.
 - Personalized product recommendations.
- **Pain Points:**
 - Frustrated by slow manual billing.
 - Wants smarter shopping experience.
- **Needs:**
 - Digital payment integration.
 - Cloud-based bill tracking.
 - Data-driven offers and recommendations.

PROPOSED SOLUTION

Key Features

1. RFID-Based Automatic Billing

- Each product carries an RFID tag.
- The trolley's RFID reader detects items as they are added or removed.
- Real-time updates are shown on an LCD display and synced to a cloud database.

2. Real-Time Bill Tracking

- Customers can monitor their expenses continuously while shopping.
- Helps with budgeting and prevents bill shock at checkout.

3. Digital Payment Integration

- Enables contactless payment directly from the trolley.
- Reduces dependency on billing counters and speeds up checkout.

4. Scalability & Cost Efficiency

- Designed to be low-cost and energy-efficient.
- Suitable for small and medium-sized retailers as well as large supermarkets.

Benefits

- **For Customers:** Faster checkout, transparent billing, easy navigation, and improved shopping experience.
- **For Stores:** Reduced staffing costs, better inventory control, and valuable customer insights.
- **For Sustainability:** Digital receipts reduce paper waste, and energy-efficient hardware lowers operational costs.

FUNCTIONAL REQUIREMENTS

1. Item Detection & Billing

- The trolley **must detect** products placed inside using RFID tags.
- It **must update** the bill automatically when items are added or removed.
- It **must display** product details (name, price, quantity) on the LCD screen.

2. Real-Time Bill Tracking

- The system **must continuously update** the total bill as items are scanned.
- Customers **must be able to view** their current bill at any time during shopping.

3. Cloud Integration

- The trolley **must sync** billing data with a cloud database.
- Store managers **must receive** real-time inventory updates.

4. Checkout Process

- The trolley **must send** the final bill to the billing server at checkout.
- Customers **must be able to complete** payment digitally (contactless option).

5. Accessibility

- The trolley **must provide** a simple, user-friendly interface.
- It **must support** elderly customers (lightweight design).

6. Multi-Trolley Operation

- The system **must handle** multiple trolleys operating simultaneously without data conflicts.

7. Error Handling

- The trolley **must store** data locally in case of network failure.

- It **must sync** with the cloud once connectivity is restored.

NON FUNCTIONAL REQUIREMENTS

1. Performance

- The system should process RFID scans in **real-time** (≤ 1 second per item).
- Billing updates must be displayed instantly on the LCD screen.
- Cloud synchronization should occur with minimal latency (< 3 seconds).

3. Usability

- Interface must be simple and intuitive for all age groups.
- large-font display should support elderly users.
- Trolley design must be lightweight and easy to maneuver.

4. Scalability

- System should support deployment in both small retail stores and large supermarkets.
- Cloud database must handle thousands of transactions simultaneously.

5. Security

- Data transmission between trolley and cloud must be encrypted.
- Customer payment data must comply with secure transaction standards.

6. Maintainability

- Hardware modules should be modular for easy replacement/upgrade.
- Software should allow updates without disrupting ongoing operations.

7. Energy Efficiency

- Trolley must use low-power microcontrollers and rechargeable batteries.
- Idle mode should reduce power consumption when not in use.

8. Cost Effectiveness

- Prototype should be built using affordable components.
- Design must minimize infrastructure investment for small/medium retailers.

PROJECT PLANNING(GANTT CHART/TIMELINE)

Phase	Duration	Key Tasks	Deliverables
1. Research & Requirement Analysis	Week 1	Study BLE, RFID, ESP32, QR payment systems; define objectives	Project proposal & component list
2. Hardware Setup	Week 2–3	Procure ESP32, RFID reader, BLE beacons, LCD, power supply	Working hardware assembly
3. BLE Integration	Week 4	Configure ESP32 as scanner & beacon; test aisle detection	BLE beacon detection demo
4. RFID Billing System	Week 5	Connect RFID reader; map product IDs to database	Automatic billing prototype
5. QR Payment Integration	Week 6	Generate QR code for UPI payment; display on LCD	Working QR payment system
6. Cloud Database Setup	Week 7	Connect ESP32 to Firebase; sync billing data	Real-time data storage
7. System Integration & Testing	Week 8–9	Combine BLE, RFID, and payment modules; debug	Fully integrated smart trolley
8. Documentation & Presentation	Week 10	Prepare report, diagrams, and viva slides	Final project submission

RISK ANALYSIS

Risk Type	Description	Impact	Mitigation Strategy
Hardware Failure	RFID reader or ESP32 may malfunction due to wiring or power issues.	High	Use voltage regulators, test connections regularly, and keep spare modules.
Signal Interference	BLE or RFID signals may overlap or weaken in crowded areas.	Medium	Adjust beacon power levels, use unique frequencies, and test placement.

Risk Type	Description	Impact	Mitigation Strategy
Data Loss / Cloud Sync Error	Firestore or database may fail to update bills or payment status.	High	Implement local backup storage and retry mechanisms for data sync.
Security Risk	QR payment data or Wi-Fi connection could be intercepted.	High	Use encrypted UPI links and secure Wi-Fi protocols (WPA2).
Tag Collision	Multiple RFID tags may be read simultaneously, causing billing errors.	Medium	Use anti-collision algorithms and limit reader range.
Battery Drain	Continuous BLE scanning and LCD display may consume power quickly.	Medium	Use power-efficient coding and rechargeable battery packs.
User Misuse	Customer may remove items without scanning or damage components.	Low	Add exit gate verification and physical protection for sensors.
Environmental Factors	Heat, dust, or moisture may affect sensors and electronics.	Medium	Use protective casing and regular maintenance.